Python Basic Assingment 6 (Class)

September 30, 2021

1 Python Basic - Assingment 6 (Class)

1.1 Problem Description

The program takes the length and breadth from the user and finds the area of the rectangle using classes.

1.2 Program Explanation

1. A class called cal is created and the **init**() method is used to initialize values of that class.
2. Methods for adding, substracting, multiplying, dividing two numbers and returning their respective results is defined.
3. The menu is printed and the choice is taken from the user.
4. An object for the class is created with the two numbers taken from the user passed as parameters.
5. Using the object, the respective method is called according to the choice taken from the user.
6. When the choice is 0, the loop is exited.
7. The final result is printed.

1.3 Problem Solution

1. Create a class and using a constructor to initialize values of that class.
2. Create methods for adding, substracting, multiplying and dividing two numbers and returning the respective results.
3. Take the two numbers as inputs and create an object for the class passing the two numbers as parameters to the class.
4. Using the object, call the respective function depending on the choice taken from the user.
5. Print the final result.
6. Exit

[ ]:
```
## Make your solutions cell b
```

[6]:
```
class cal:
  def __init__(self, n1, n2):
    self.n1 = n1
    self.n2 = n2

  def add(ad):
    return ad.n1+ad.n2
```

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  def sub(su):
    return su.n1-su.n2

  def mul(mu):
    return mu.n1*mu.n2

  def div(di):
    return di.n1/di.n2

x=[int(x) for x in input("Enter both numbers by putting space: ").split()]
objct_creation = cal(x[0],x[1])
choice = True
while choice != 0:
    print("Choose your option")
    print(" 1. Add \n 2. Subtraction \n 3. Multiplication \n 4. Division \n 0. Exit" )

    choice=int(input("Enter your choice: "))

    if choice==1:
        print("Result: ",objct_creation.add())
    elif choice==2:
        print("Result: ",objct_creation.sub())
    elif choice==3:
        print("Result: ",objct_creation.mul())
    elif choice==4:
        print("Result: ",round(objct_creation.div(),3))
    elif choice==0:
        print("Stop calculation")
    else:
        print("wrong option")
```

```
Enter both numbers by putting space:  10 5

Choose your option
 1. Add
 2. Subtraction
 3. Multiplication
 4. Division
 0. Exit

Enter your choice:  4

Result:  2.0
Choose your option
 1. Add
 2. Subtraction
 3. Multiplication
```

```
 4. Division
 0. Exit

Enter your choice:  0

Stop calculation
```

1.3.1 Example solutions Assingment 6.

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```
class cal():
    def __init__(self,a,b):
        self.a=a
        self.b=b
    def add(self):
        return self.a+self.b
    def mul(self):
        return self.a*self.b
    def div(self):
        return self.a/self.b
    def sub(self):
        return self.a-self.b
a=int(input("Enter first number: "))
b=int(input("Enter second number: "))
obj=cal(a,b)
choice=1
while choice!=0:
    print("0. Exit")
    print("1. Add")
    print("2. Subtraction")
    print("3. Multiplication")
    print("4. Division")
    choice=int(input("Enter choice: "))
    if choice==1:
        print("Result: ",obj.add())
    elif choice==2:
        print("Result: ",obj.sub())
    elif choice==3:
        print("Result: ",obj.mul())
    elif choice==4:
        print("Result: ",round(obj.div(),2))
    elif choice==0:
        print("Exiting!")
    else:
        print("Invalid choice!!")


print()
```

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